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Spring-Data-JPA-Hibernate

Difference Between JPA, Hibernate, and Spring Data JPA

Java Persistence API (JPA)

- JPA is a Java specification (JSR 338) for persisting, reading, and managing data from Java objects.
- It does not contain any concrete implementation.
- It is a set of interfaces and annotations.

Hibernate

- Hibernate is an ORM (Object-Relational Mapping) tool that implements the JPA specification.
- It provides concrete implementation of all JPA interfaces.
- It manages database operations and SQL generation.

Spring Data JPA

- Spring Data JPA does not have its own JPA implementation.
- It reduces boilerplate code required to implement data access layers.
- It is another level of abstraction over JPA implementation providers like Hibernate.
- It automatically manages transactions.

Exercise 2: Spring Data JPA - Quick Example

Objective

Create a Spring Boot application using Spring Data JPA and Hibernate to manage country data in a MySQL database.

Steps

1. A Spring Boot project is created using start.spring.io with the following configurations:
 - Group: com.cognizant
 - Artifact: orm-learn
 - Description: "Demo project for Spring Data JPA and Hibernate"
 - Dependencies: Spring Boot DevTools, Spring Data JPA, MySQL Driver
2. A new schema ormlearn is created in MySQL database.
3. The application.properties file is configured with database connection details and logging configuration.
4. The country table is created in MySQL with columns for code and name, and sample records are inserted.
5. A package com.cognizant.ormlearn.model is created with Country.java entity class using @Entity, @Table, @Id, and @Column annotations.
6. A package com.cognizant.ormlearn.repository is created with CountryRepository interface extending JpaRepository<Country, String> and @Repository annotation.
7. A package com.cognizant.ormlearn.service is created with CountryService class using @Service annotation, autowired CountryRepository, and a getAllCountries() method with @Transactional.
8. The OrmLearnApplication main class is modified to load the Spring context, get the CountryService bean, and call testGetAllCountries() method.
9. The application is executed and verified that country data is retrieved from the database.

Key Concepts

- @Entity marks a class as a JPA entity that maps to a database table
- @Table(name="country") specifies the table name in the database
- @Id marks the primary key field
- @Column(name="co_code") maps a field to a specific database column
- JpaRepository<Country, String> provides built-in CRUD operations
- @Repository marks an interface as a Spring Data repository
- @Service marks a class as a service layer component
- @Transactional ensures the method runs within a transaction boundary
- @Autowired injects the repository dependency automatically
- Spring Data JPA automatically implements the repository interface at runtime

Output

The application displays the list of countries retrieved from the MySQL database:
countries=[Country [code=IN, name=India], Country [code=US, name=United States of America]]

Output:

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:: Spring Boot ::
(v2.7.18)

02-07-26 23:20:06.899 restartedMain INFO c.c.o.OrmLearnApplication logStarting 55 Starting OrmLearnApplication using Java 26.0.1 on ADARSHs-MacBook-Air-2.1
ocal with PID 26927 (/Users/adarsh/Developer/github/genc-dn5.0/week2/spring-jpa-hibernate/target/classes started by adarsh in /Users/adarsh/Developer/github/genc-dn5.0/week2/sprin
g-jpa-hibernate)
02-07-26 23:20:06.899 restartedMain DEBUG c.c.o.OrmLearnApplication logStarting 56 Running with Spring Boot v2.7.18, Spring v5.3.31
02-07-26 23:20:06.900 restartedMain INFO c.c.o.OrmLearnApplication logStartupProfileInfo 632 No active profile set, falling back to 1 default profile: "default"
02-07-26 23:20:06.931 restartedMain INFO ertyDefaultsPostProcessor logTo 255 Devtools property defaults active! Set 'spring.devtools.add-properties' t
o 'false' to disable
02-07-26 23:20:07.163 restartedMain INFO toryConfigurationDelegate registerRepositoriesIn 132 Bootstrapping Spring Data JPA repositories in DEFAULT mode.
02-07-26 23:20:07.189 restartedMain INFO toryConfigurationDelegate registerRepositoriesIn 201 Finished Spring Data repository scanning in 21 ms. Found 1 JPA repository
interfaces.
02-07-26 23:20:07.425 restartedMain INFO o.h.j.i.util.LogHelper logPersistenceUnitInformation 31 HHH000204: Processing PersistenceUnitInfo [name: default]
02-07-26 23:20:07.452 restartedMain INFO org.hibernate.Version logVersion 44 HHH000412: Hibernate ORM core version 5.6.15.Final
02-07-26 23:20:07.544 restartedMain INFO o.h.a.common.Version <clinit> 56 HCANN000001: Hibernate Commons Annotations {5.1.2.Final}
02-07-26 23:20:07.603 restartedMain INFO c.z.h.HikariDataSource getConnection 110 HikariPool-1 - Starting...
02-07-26 23:20:07.901 restartedMain INFO c.z.h.HikariDataSource getConnection 123 HikariPool-1 - Start completed.
02-07-26 23:20:07.910 restartedMain INFO o.h.dialect.Dialect <init> 175 HHH000400: Using dialect: org.hibernate.dialect.MySQL5Dialect
02-07-26 23:20:08.222 restartedMain INFO .p.i.JtaPlatformInitiator initiateService 52 HHH000490: Using JtaPlatform implementation: [org.hibernate.engine.transa
ction.jta.platform.internal.NoJtaPlatform]
02-07-26 23:20:08.227 restartedMain INFO rEntityManagerFactoryBean buildNativeEntityManagerFactory 437 Initialized JPA EntityManagerFactory for persistence unit 'default'
02-07-26 23:20:08.423 restartedMain INFO .OptionalLiveReloadServer startServer 59 LiveReload server is running on port 35729
02-07-26 23:20:08.434 restartedMain INFO c.c.o.OrmLearnApplication logStarted 61 Started OrmLearnApplication in 1.756 seconds (JVM running for 2.024)
02-07-26 23:20:08.436 restartedMain INFO c.c.o.OrmLearnApplication main 22 Inside main
02-07-26 23:20:08.436 restartedMain INFO c.c.o.OrmLearnApplication testGetAllCountries 27 Start
02-07-26 23:20:08.508 restartedMain DEBUG org.hibernate.SQL logStatement 144 select country0_._co_code as co_code1_0_, country0_._co_name as co_name2_0_
from country0_
02-07-26 23:20:08.513 restartedMain TRACE o.h.t.d.s.BasicExtractor extract 60 extracted value ([co_code1_0_] : [VARCHAR]) - [IN]
02-07-26 23:20:08.516 restartedMain TRACE o.h.t.d.s.BasicExtractor extract 60 extracted value ([co_name2_0_] : [VARCHAR]) - [India]
02-07-26 23:20:08.516 restartedMain TRACE o.h.t.d.s.BasicExtractor extract 60 extracted value ([co_code1_0_] : [VARCHAR]) - [US]
02-07-26 23:20:08.517 restartedMain TRACE o.h.t.d.s.BasicExtractor extract 60 extracted value ([co_name2_0_] : [VARCHAR]) - [United States of America]
02-07-26 23:20:08.526 restartedMain DEBUG c.c.o.OrmLearnApplication testGetAllCountries 29 countries=[Country [code=IN, name=India], Country [code=US, name=United S
tates of America]]
02-07-26 23:20:08.526 restartedMain INFO c.c.o.OrmLearnApplication testGetAllCountries 30 End
02-07-26 23:20:08.531 locationShutdownHook INFO rEntityManagerFactoryBean destroy 651 Closing JPA EntityManagerFactory for persistence unit 'default'
02-07-26 23:20:08.532 locationShutdownHook INFO c.z.h.HikariDataSource close 350 HikariPool-1 - Shutdown initiated...
02-07-26 23:20:08.536 locationShutdownHook INFO c.z.h.HikariDataSource close 352 HikariPool-1 - Shutdown completed.

[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 4.082 s
[INFO] Finished at: 2026-07-02T23:20:08+05:30
[INFO]
```